

A

Project Stage-I

Report

On

**“AI Powered Meeting Summarizer
and Action Item Generator”**

By

1. Patil Harshal Mahesh (TY AIML-231107040)
2. Patil Diksha Sudhakar (TY AIML-231107026)
3. Pardeshi Tushar Rajendra (TY AIML-231107037)
4. Patil Harshal Dnyaneshwar (TY AIML-231107057)



R. C. PATEL
INSTITUTE OF TECHNOLOGY
An Autonomous Institute

**Department of Artificial Intelligence and
Machine Learning**

The Shirpur Education Society's

R. C. Patel Institute of Technology, Shirpur - 425405.

[2025-26]

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Action Item Generator”

In partial fulfillment of requirements for the degree of
Bachelor of Technology

In
Artificial Intelligence and Machine Learning

Submitted By

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Under the Guidance of

Prof. Dr. U M Patil



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Artificial Intelligence and Machine Learning

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INSTITUTE OF TECHNOLOGY**
An Autonomous Institute

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE
LEARNING**

Institute Vision:

Fostering technical excellence through ethics, sustainable education, innovation and research.

Institute Mission:

To impart high quality Technical Education through:

- M1.** Innovative and Interactive learning process and high quality, globally recognized instructional programs.
- M2.** Fostering a collaborative scientific temper among students with ethical responsibility towards the society.
- M3.** Preparing students from diverse backgrounds to have Aptitude for employment, entrepreneurship and research with a spirit of Professionalism.
- M4.** To contribute to nation's sustainable development.

Department Vision:

Fostering technical excellence in Artificial Intelligence and Machine Learning through ethics, innovative research, and sustainable education.

Department Mission:

- M1.** Innovative and interactive learning processes with high quality instructional programs in Artificial Intelligence and Machine Learning.
- M2.** Fostering a collaborative scientific temper among students with ethical responsibility towards society in AI and ML Applications.
- M3.** Preparing students from diverse backgrounds to develop Aptitude for employment, entrepreneurship, and research in AI and ML with a spirit of professionalism.



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Artificial Intelligence and Machine Learning *CERTIFICATE*

This is to certify that the Project Stage-I entitled “**AI Powered Meeting Summarizer and Action Item Generator**” has been carried out by team:

- 1. Patil Harshal Mahesh**
- 2. Patil Diksha Sudhakar**
- 3. Pardeshi Tushar Rajendra**
- 4. Patil Harshal Dnyaneshwar**

under the guidance of **Prof. Dr. U M Patil** in partial fulfillment of the requirement for the degree of Bachelor of Technology in Department of Artificial Intelligence and Machine Learning (Semester-VI) of Dr. Babasaheb Ambedkar Technological University, Lonere during the academic year 2025-26.

Date:

Place: Shirpur

Guide

Prof. Dr. U M Patil

Project Stage-I Coordinator

Prof. P. D. Lanjewar

H.O.D.

Prof. Dr. U. M. Patil

Director

Prof. Dr. J. B. Patil

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Project Team:

1. Patil Harshal Mahesh
2. Patil Diksha Sudhakar
3. Pardeshi Tushar Rajendra
4. Patil Harshal Dnyaneshwar

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ABSTRACT

Meetings are an essential part of organizational communication and decision-making. However, manually preparing meeting summaries, tracking decisions, and identifying action items is a time-consuming process that may lead to missing important information. To address this challenge, the **AI Powered Meeting Summarizer and Action Item Generator** has been developed using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques.

The proposed system analyzes meeting transcripts and automatically generates concise summaries, extracts important decisions, identifies action items, detects risks, and performs sentiment analysis. It also calculates productivity scores and provides analytical insights through interactive dashboards. The system includes a knowledge base for storing and retrieving historical meeting records and a report generation module for creating structured meeting reports.

The project is implemented using Python, Streamlit, spaCy, NLTK, Scikit-Learn, SQLite, and Plotly. By automating meeting documentation and analysis, the system reduces manual effort, improves information accessibility, and enhances organizational productivity. The developed solution demonstrates the practical application of AI and NLP technologies in intelligent meeting management and knowledge processing.

CHAPTER – 1

INTRODUCTION

1. INTRODUCTION

Meetings play an important role in organizations, educational institutions, and businesses for discussing ideas, making decisions, assigning tasks, and planning future activities. However, manually recording meeting discussions, preparing summaries, identifying important decisions, and tracking action items can be time-consuming and prone to human errors. As the number of meetings increases, managing meeting records and extracting useful information becomes increasingly difficult.

Recent advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) have made it possible to automatically analyze textual information and generate meaningful insights. AI-powered summarization techniques can convert lengthy meeting transcripts into concise summaries, while NLP algorithms can identify important decisions, action items, risks, discussion topics, and participant contributions.

The proposed project, **“AI Powered Meeting Summarizer and Action Item Generator,”** is designed to automate the analysis of meeting transcripts. The system processes meeting conversations, generates multi-level summaries, extracts decisions and action items, performs sentiment analysis, identifies risks, and provides productivity scores. It also maintains a searchable knowledge base of meetings and generates detailed reports for future reference.

1.1. Problem Statement

In many organizations, meetings generate large amounts of information in the form of discussions, decisions, tasks, and recommendations. Traditionally, meeting minutes are prepared manually, which requires significant time and effort. Manual documentation may result in missing important information, inaccurate summaries, or overlooked action items.

Additionally, organizations often struggle to track decisions made during meetings and ensure that assigned tasks are completed on time. Retrieving information from previous meetings also becomes difficult when records are not properly organized.

Therefore, there is a need for an intelligent system that can automatically analyze meeting transcripts, generate summaries, identify action items, detect risks, and store meeting knowledge in an organized manner. Such a system can improve productivity, reduce manual effort, and enhance decision-making efficiency.

1.2. Objectives

The main objectives of the proposed system are as follows:

1.2.1. Automatic Meeting Summarization

To generate concise and meaningful summaries from lengthy meeting transcripts using Natural Language Processing techniques.

1.2.2. Action Item Extraction

To automatically identify tasks, responsibilities, deadlines, and action items discussed during meetings.

1.2.3. Decision Detection

To detect and extract important decisions taken during meetings for future reference.

1.2.4. Risk Identification

To identify potential risks, concerns, and challenges discussed by participants.

1.2.5. Sentiment Analysis

To analyze participant sentiments and determine positive, negative, or neutral opinions expressed during meetings.

1.2.6. Productivity Measurement

To calculate meeting productivity scores based on decisions, actions, participation, and discussion quality.

1.3. Application

The proposed system can be applied in various domains where meeting management and documentation are important.

1.3.1. Corporate Organizations

The system can automatically generate meeting minutes, track decisions, and monitor assigned tasks in business organizations.

1.3.2. Educational Institutions

Colleges and universities can use the system to document academic meetings, project discussions, and administrative activities.

1.3.3. Project Management

Project managers can use the system to track project discussions, responsibilities, deadlines, and progress updates.

1.3.4. Government and Administrative Offices

Government departments can maintain structured records of official meetings and policy discussions.

1.3.5. Research and Development Teams

Research groups can organize meeting discussions, experimental findings, and future action plans efficiently.

CHAPTER – 2

LITERATURE SURVEY

2. LITERATURE SURVEY

The increasing use of online and offline meetings in organizations has created a need for automated systems that can analyze discussions, summarize important information, and identify action items. Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning techniques have been widely applied to automate meeting analysis and improve productivity. Various research studies have proposed methods for meeting summarization, decision extraction, sentiment analysis, and task identification. This literature survey reviews existing research work related to meeting summarization systems and intelligent meeting assistants.

2.1. Background

Meetings generate a large amount of textual information in the form of discussions, decisions, suggestions, and tasks. Traditionally, meeting minutes are prepared manually, which requires considerable effort and may result in incomplete documentation. To overcome these challenges, researchers have explored NLP-based approaches for automatic summarization and information extraction.

Natural Language Processing enables computers to understand, process, and analyze human language. Techniques such as TF-IDF, TextRank, sentence ranking, keyword extraction, and transformer-based language models have significantly improved automatic summarization systems. In recent years,

Large Language Models (LLMs) have further enhanced the quality of generated summaries and action-item extraction.

Automated meeting analysis systems help organizations improve efficiency by reducing manual work, tracking decisions, identifying responsibilities, and maintaining searchable knowledge repositories.

2.2. Existing Systems

Several researchers and organizations have developed systems for automatic meeting summarization and meeting intelligence. The following studies were reviewed during the development of the proposed system.

2.2.1. Automatic Text Summarization using TF-IDF

Ramos proposed the use of Term Frequency–Inverse Document Frequency (TF-IDF) for identifying important words and sentences within textual documents. TF-IDF became one of the most widely used techniques for extractive text summarization and keyword extraction.

Citation:

[1] J. Ramos, “Using TF-IDF to Determine Word Relevance in Document Queries,” Proceedings of the First Instructional Conference on Machine Learning, 2003.

2.2.2. TextRank Based Summarization

Mihalcea and Tarau introduced the TextRank algorithm, a graph-based ranking method for text summarization and keyword extraction. The approach ranks sentences based on importance and generates concise summaries from large documents.

Citation:

[2] R. Mihalcea and P. Tarau, “TextRank: Bringing Order into Text,” Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2004.

2.2.3. Meeting Summarization using Natural Language Processing

Murray, Renals, and Carletta studied automatic meeting summarization techniques and demonstrated how NLP methods can generate summaries from spoken meeting transcripts. Their work highlighted the importance of extracting key discussion points and decisions from meetings.

Citation:

[3] G. Murray, S. Renals, and J. Carletta, "Extractive Summarization of Meeting Recordings," Proceedings of Interspeech Conference, 2005.

2.2.4. Transformer-Based Language Models

Devlin et al. introduced BERT (Bidirectional Encoder Representations from Transformers), which significantly improved language understanding and text summarization tasks. Transformer-based models have become the foundation for modern AI-powered summarization systems.

Citation:

[4] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," Proceedings of NAACL-HLT, 2019.

2.3. Comparative Study of Existing Systems

Research Work	Technology Used	Advantages	Limitations
Ramos (2003)	TF-IDF	Simple and efficient keyword extraction	Limited contextual understanding
Mihalcea & Tarau (2004)	TextRank	Effective sentence ranking	May miss semantic meaning
Murray et al. (2005)	Meeting Summarization	Extracts key discussion points	Limited scalability
Devlin et al. (2019)	BERT	Better contextual understanding	High computational cost
Touvron et al. (2023)	LLaMA	Human-like summary generation	Requires significant resources

CHAPTER – 3

METHODOLOGY

3. METHODOLOGY

The methodology describes the overall approach used to design and implement the AI Powered Meeting Summarizer and Action Item Generator. The system uses Natural Language Processing (NLP), Artificial Intelligence (AI), and optional Large Language Models (LLMs) to analyze meeting transcripts and generate meaningful outputs such as summaries, decisions, action items, risks, and productivity scores.

The methodology consists of hardware and software requirements, system architecture, algorithm design, dataset analysis, and visualization techniques.

3.1. Hardware and Software Requirement

3.1.1. Hardware Requirements

Component	Specification
Processor	Intel Core i3/i5/i7 or higher
RAM	8 GB or Above
Storage	50 GB Free Space
Internet Connection	Required for LLM Integration
Display	Standard Monitor
Input Devices	Keyboard and Mouse

Table 3.1: Hardware Requirements

The system can run on a standard personal computer without requiring specialized hardware.

3.1.2. Software Requirements

Software	Purpose
Windows 10/11	Operating System
Python 3.x	Programming Language
Streamlit	Web Application Framework
spaCy	NLP Processing
NLTK	Text Processing
Scikit-Learn	TF-IDF Analysis
SQLite	Database Management
Plotly	Data Visualization
Ollama	LLM Integration
Visual Studio Code	Development Environment

Table 3.2: Software Requirements

These software tools work together to process meeting transcripts and generate analytical insights.

3.2. System Design

The proposed system follows an AI-driven workflow where meeting transcripts are analyzed using Natural Language Processing techniques. The extracted information is then organized and presented through dashboards, reports, and knowledge-base modules.

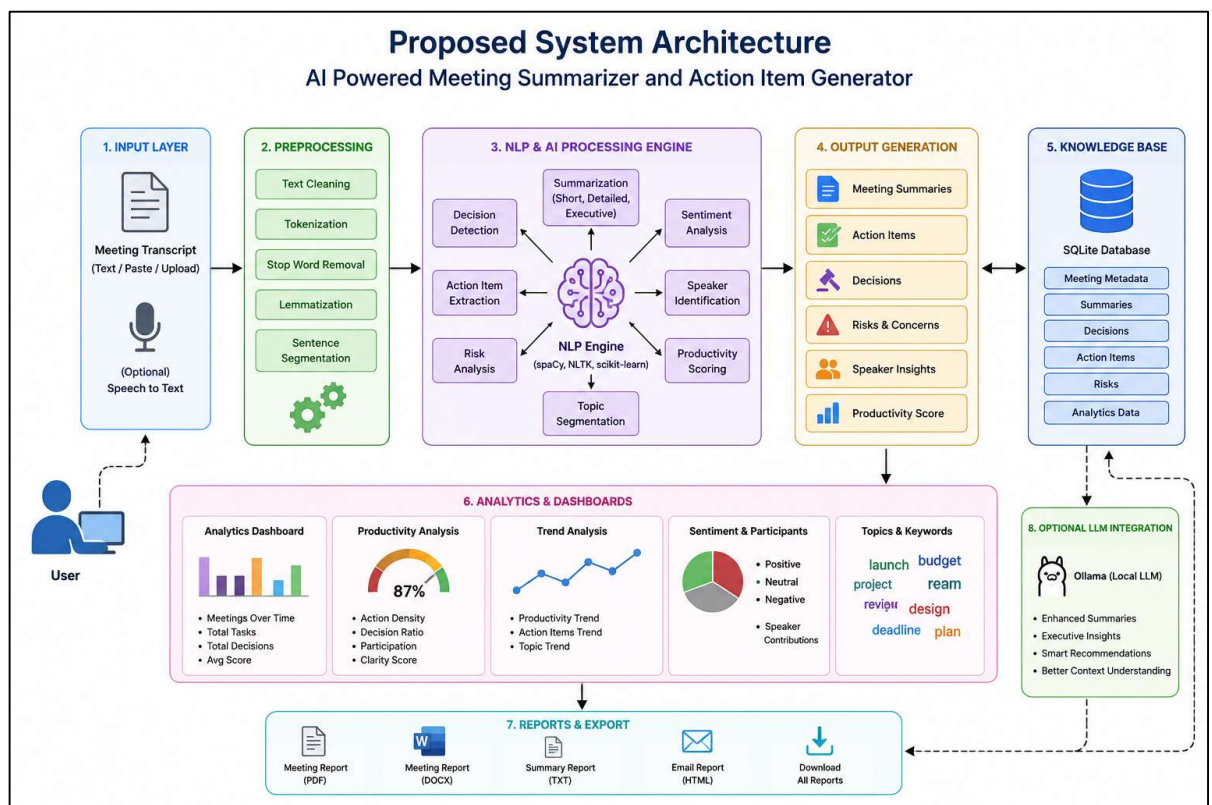


Figure 3.1: Proposed System Architecture

Working of System Design

1. User enters meeting transcript.
2. NLP engine preprocesses the text.
3. Important keywords are extracted.
4. Summary generation module creates short and detailed summaries.

5. Action item extraction module identifies tasks and responsibilities.
6. Decision detection module extracts important decisions.
7. Risk analysis module identifies concerns and issues.
8. Sentiment analysis module evaluates participant opinions.
9. Analytics module generates productivity insights.
10. Knowledge base stores meeting records.
11. Report generator creates downloadable reports. The architecture ensures secure and automated voter verification.

The architecture ensures automated meeting intelligence and efficient knowledge management.

3.3. Algorithm

3.3.1. TF-IDF Algorithm

TF-IDF (Term Frequency-Inverse Document Frequency) is used to identify important keywords and sentences from meeting transcripts.

- **Steps**

1. Read meeting transcript.
2. Tokenize text into words.
3. Calculate term frequency.
4. Calculate inverse document frequency.
5. Compute TF-IDF score.
6. Select important keywords and sentences.

- **Advantages**

- Simple implementation
- Fast processing
- Effective keyword extraction

3.3.2. NLP-Based Summarization Algorithm

The summarization module generates concise summaries by ranking important sentences.

- **Steps**

1. Preprocess transcript.
2. Remove stop words.
3. Extract keywords.
4. Rank sentences.
5. Generate summary

- **Advantages**

- Reduces lengthy text
- Preserves important information
- Improves readability

3.3.4. Action Item Extraction Algorithm

The action-item extraction module identifies tasks assigned during meetings.

- **Steps**

1. Analyze transcript.
2. Detect task-related phrases.
3. Identify responsible participants.
4. Extract deadlines if available.
5. Generate task list.

- **Advantages**

- Automates task tracking
- Improves accountability
- Reduces manual effort

3.4. Exploratory Data Analysis and Dataset Visualization

3.4.1. Dataset Used

The project uses a Meeting Transcript Dataset.

Dataset Components:

- Meeting Title
- Meeting Date
- Participants
- Discussion Transcript
- Decisions
- Action Items
- Risks
- Meeting Duration

3.4.2. Dataset Features

Feature	Description
Meeting Title	Name of Meeting
Transcript	Complete Discussion Text
Participants	Meeting Members
Decisions	Important Decisions
Action Items	Assigned Tasks
Risks	Identified Concerns
Duration	Meeting Length
Productivity Score	Meeting Effectiveness

Table 3.3: Dataset Features

CHAPTER – 4

IMPLEMENTATION DETAILS

4. IMPLEMENTATION DETAILS

The implementation phase describes the practical development of the proposed system. The **“AI Powered Meeting Summarizer and Action Item Generator”** is implemented using Python, Streamlit, Natural Language Processing (NLP), SQLite Database, and Artificial Intelligence technologies. The system analyzes meeting transcripts, generates summaries, extracts decisions and action items, identifies risks, performs sentiment analysis, and generates productivity reports.

The implementation is divided into different modules, where each module performs a specific task in the meeting analysis process.

4.1. Dataset Used

The dataset used in this project consists of meeting transcripts containing discussions, decisions, action items, participant information, and meeting details. The transcripts are analyzed using NLP techniques to generate meaningful insights and reports.

The system processes textual meeting data entered by users and stores the analyzed results in a SQLite database for future reference and knowledge management.

Dataset Information

- **Dataset Type:** Meeting Transcript Dataset
- **Source:** User Input / Sample Meeting Records
- **Data Format:** Text Documents
- **Processing Technique:** Natural Language Processing (NLP)
- **Database:** SQLite
- **Analysis Method:** TF-IDF, NLP, Sentiment Analysis

4.2. Module Wise Description

The system is divided into three major modules:

- Meeting Analysis Module
- Knowledge Base & Analytics Module
- Report Generation Module

4.2.1. Module 1 – Meeting Analysis Module

Description

The Meeting Analysis Module is the core component of the system. It processes meeting transcripts using NLP techniques and extracts important information such as summaries, decisions, action items, risks, keywords, and participant contributions.

Working

- User enters meeting transcript.
- NLP engine preprocesses the text.
- Keywords are extracted.
- Summary is generated.
- Decisions are identified.
- Action items are extracted.
- Risks and concerns are detected.
- Productivity score is calculated.

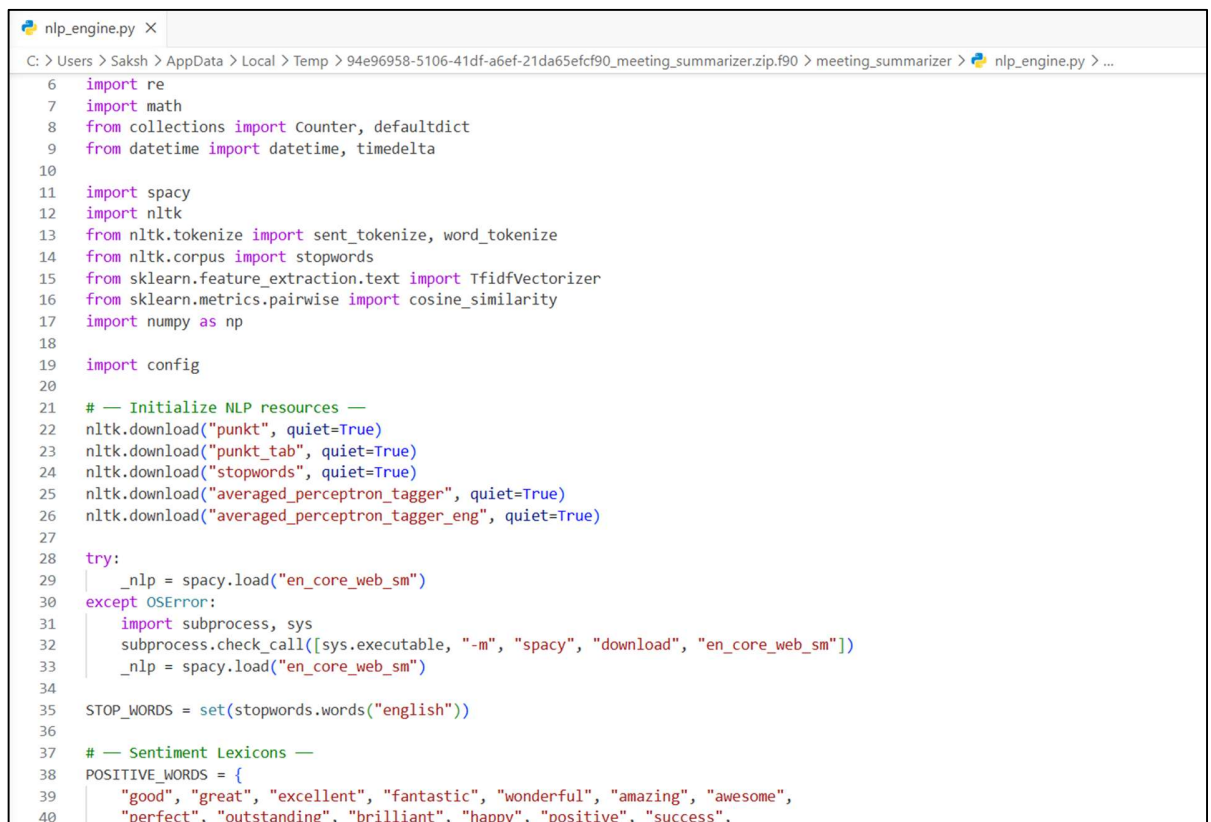
Testing

Expected Result:

- Meeting summary should be generated accurately.
- Decisions should be extracted correctly.
- Action items should be identified automatically.

Actual Result:

The module successfully generated meeting summaries, extracted decisions, identified action items, and calculated productivity scores.



```

nlp_engine.py X
C: > Users > Saksh > AppData > Local > Temp > 94e96958-5106-41df-a6ef-21da65efcf90_meeting_summarizer.zip.f90 > meeting_summarizer > nlp_engine.py > ...

6 import re
7 import math
8 from collections import Counter, defaultdict
9 from datetime import datetime, timedelta
10
11 import spacy
12 import nltk
13 from nltk.tokenize import sent_tokenize, word_tokenize
14 from nltk.corpus import stopwords
15 from sklearn.feature_extraction.text import TfidfVectorizer
16 from sklearn.metrics.pairwise import cosine_similarity
17 import numpy as np
18
19 import config
20
21 # — Initialize NLP resources —
22 nltk.download("punkt", quiet=True)
23 nltk.download("punkt_tab", quiet=True)
24 nltk.download("stopwords", quiet=True)
25 nltk.download("averaged_perceptron_tagger", quiet=True)
26 nltk.download("averaged_perceptron_tagger_eng", quiet=True)
27
28 try:
29     _nlp = spacy.load("en_core_web_sm")
30 except OSError:
31     import subprocess, sys
32     subprocess.check_call([sys.executable, "-m", "spacy", "download", "en_core_web_sm"])
33     _nlp = spacy.load("en_core_web_sm")
34
35 STOP_WORDS = set(stopwords.words("english"))
36
37 # — Sentiment Lexicons —
38 POSITIVE_WORDS = {
39     "good", "great", "excellent", "fantastic", "wonderful", "amazing", "awesome",
40     "perfect", "outstanding", "brilliant", "happy", "positive", "success",

```

Figure 4.1: NLP Processing Code

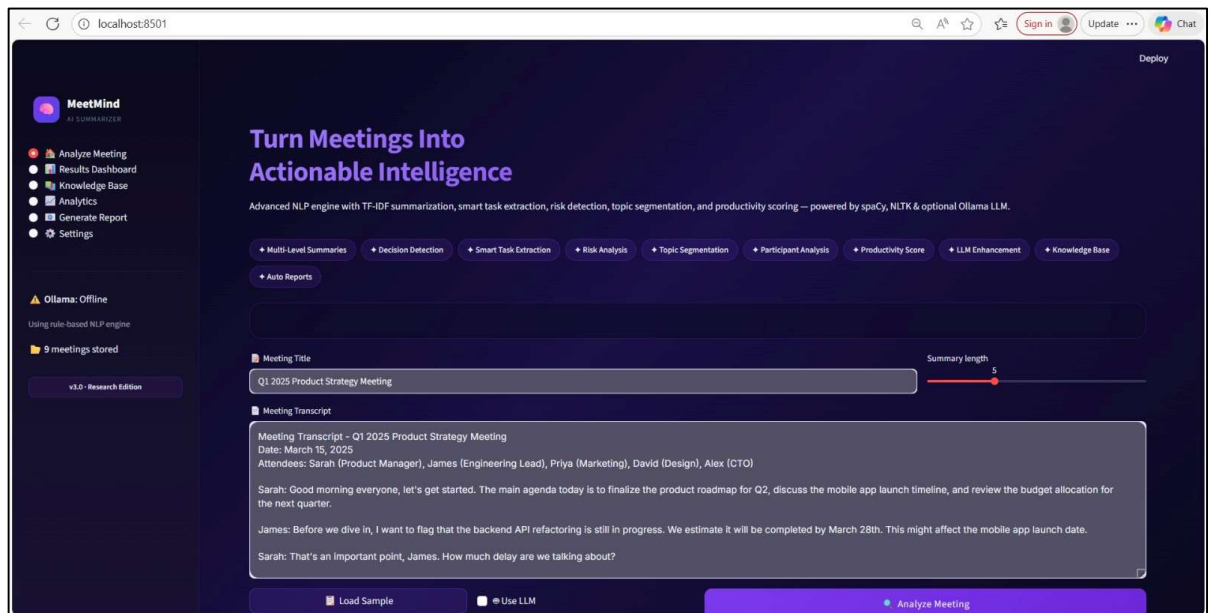


Figure 4.2: Meeting Analysis Interface

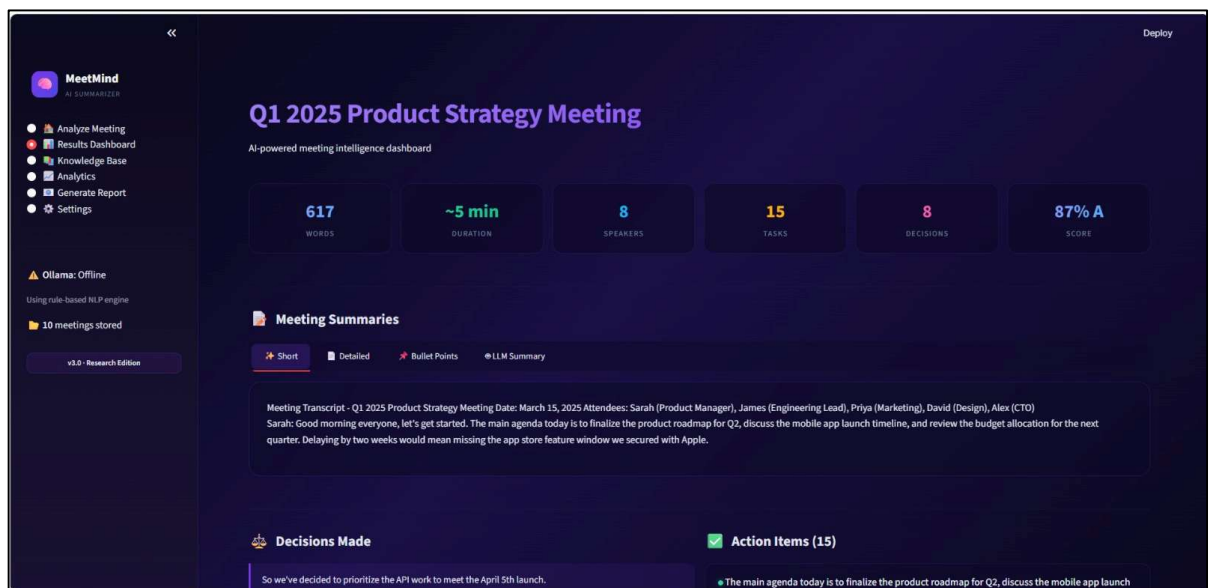


Figure 4.3: Meeting Summary and Action Item Output

4.2.2. Module 2 – Knowledge Base and Analytics Module

Description

The Knowledge Base Module stores analyzed meetings in a SQLite database and provides search functionality for retrieving historical meeting information. The Analytics Module generates productivity insights and meeting statistics using visual dashboards.

Working

- Store analyzed meetings in database.
- Save summaries and decisions.
- Store action items.
- Generate analytics dashboards.
- Calculate productivity trends.
- Retrieve historical meetings.

Testing

Expected Result:

- Meeting records should be stored successfully.
- Analytics dashboards should display accurate information.

Actual Result:

Meeting records were stored successfully, and analytics dashboards generated meaningful insights and productivity trends.

```

knowledge_base.py 1 X
C: > Users > Saksh > AppData > Local > Temp > 899797eb-7c5d-4d06-bb2b-b177876628b7_meeting_summarizer.zip.8b7 > meeting_summarizer > knowledge_base.py >

6 import json
7 import sqlite3
8 import threading
9 from datetime import datetime
10 from typing import Optional
11 import config
12
13
14 class KnowledgeBase:
15     """Persistent meeting storage with full-text search."""
16
17     def __init__(self, db_path: str = None):
18         self.db_path = db_path or config.DB_PATH
19         self._local = threading.local() # Fix #1 & #3: thread-local storage
20         self._init_db()
21
22     def _get_conn(self) -> sqlite3.Connection:
23         """Return a per-thread SQLite connection (Fix #1)."""
24         if not hasattr(self._local, "conn") or self._local.conn is None:
25             conn = sqlite3.connect(self.db_path, check_same_thread=False)
26             conn.row_factory = sqlite3.Row
27             if self.db_path != ":memory:":
28                 conn.execute("PRAGMA journal_mode=WAL")
29             self._local.conn = conn
30         return self._local.conn
31
32     def close(self):
33         """Close this thread's database connection (Fix #2)."""
34         conn = getattr(self._local, "conn", None)
35         if conn:
36             conn.close()
37             self._local.conn = None
38
39     def _init_db(self):
40         """Create tables if they don't exist."""

```

Figure 4.4: Database Management Code

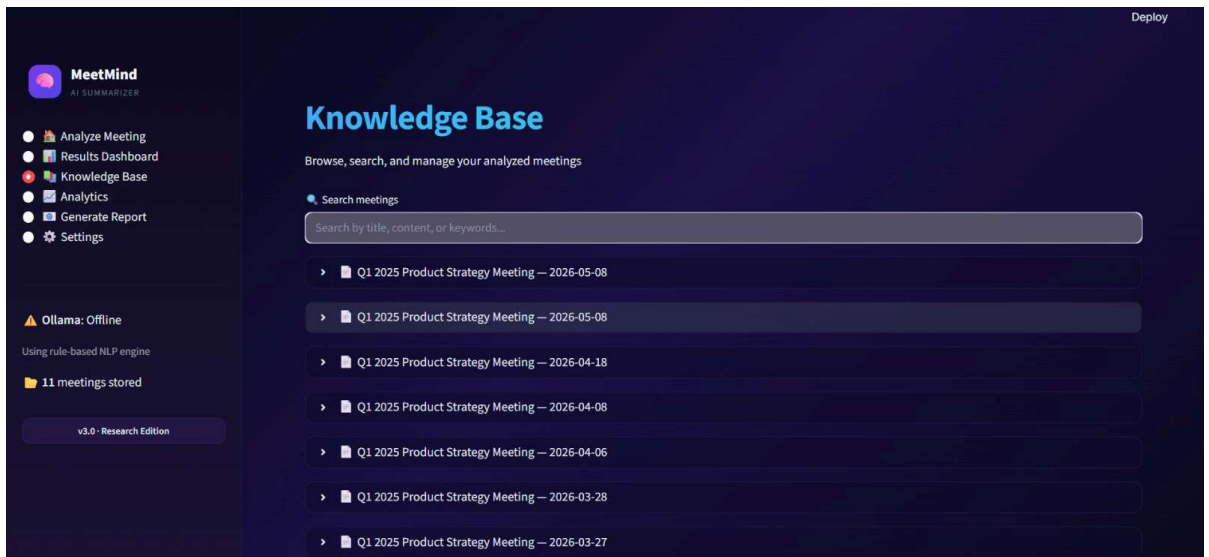


Figure 4.5: Knowledge Base Interface

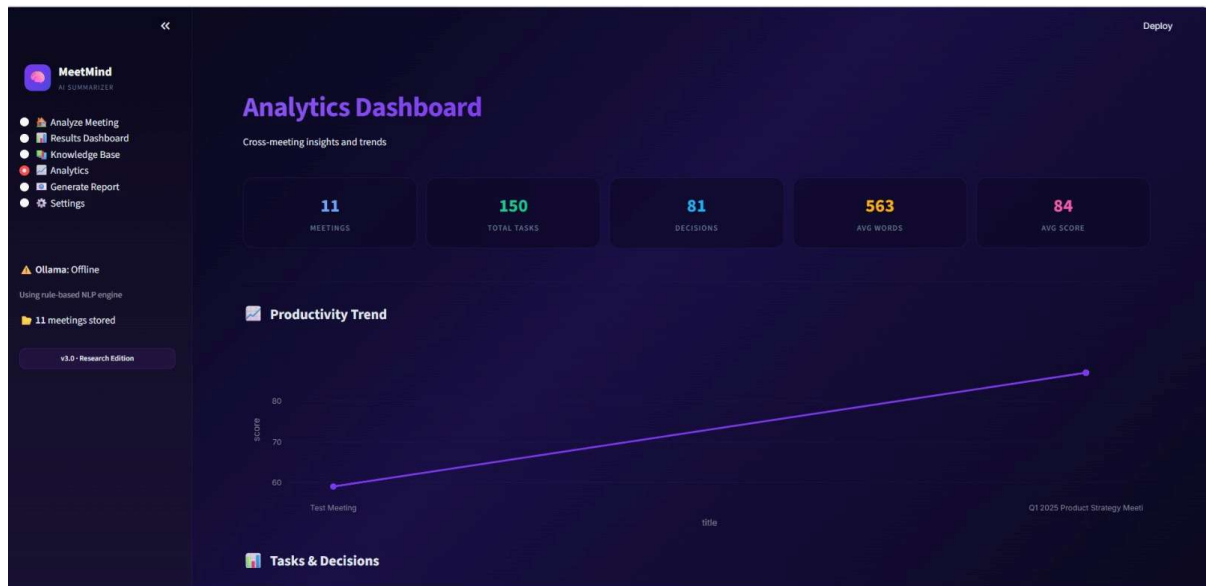


Figure 4.6: Analytics Dashboard

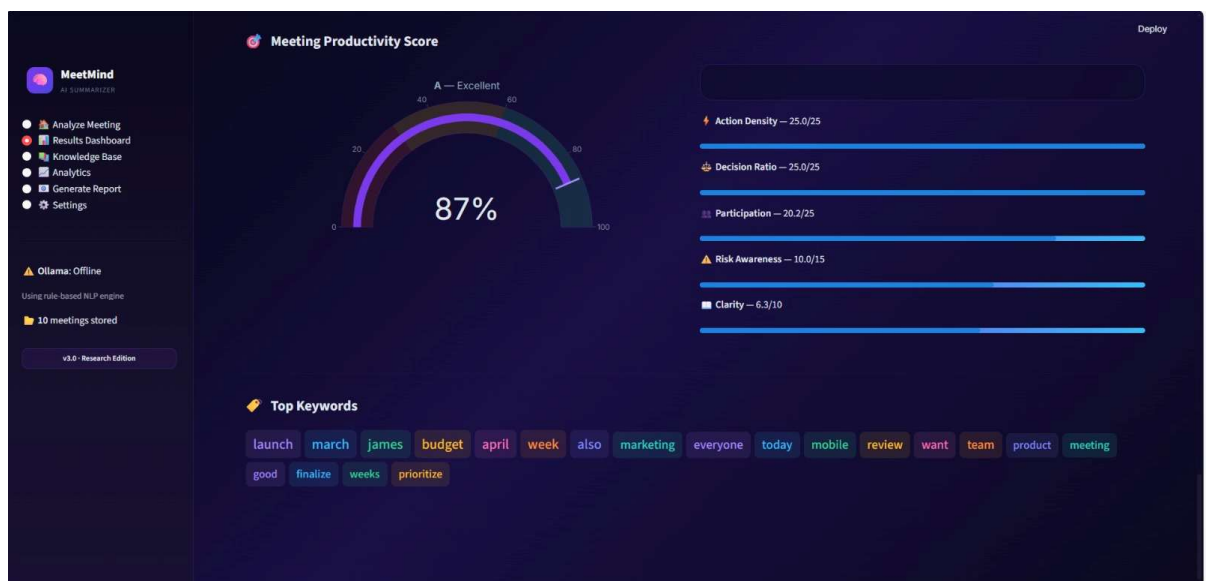


Figure 4.7: Productivity Score Dashboard

4.2.3. Module 3 – Report Generation Module

Description

The Report Generation Module creates detailed meeting reports based on the analysis results. The generated reports include summaries, decisions, action items, productivity scores, risks, and participant analysis.

Working:

- Collect meeting analysis results.
- Generate executive summary.
- Compile decisions and action items.
- Create report structure.
- Export report for download.

Testing:

Expected Result:

- Detailed reports should be generated automatically.
- Download functionality should work correctly.

Actual Result:

The module successfully generated detailed meeting reports and provided report download functionality.

```

report_generator.py 1 X
D:\> Downloads > meeting_summarizer > meeting_summarizer > report_generator.py > ...

5 from datetime import datetime
6 from jinja2 import Template
7
8
9 # 📄 Markdown Report Template 📄
10 MARKDOWN_TEMPLATE = Template("""# 📄 Meeting Report: {{ title }}
11
12 **Date:** {{ date }}
13 **Duration:** ~({{ stats.estimated_duration }} minutes | **Words:** {{ stats.word_count | default('N/A') }} | **Speakers:** {{ stats.speaker_count | default('N/A') }}
14 **Productivity Score:** {{ productivity_score.total }}/100 ({{ productivity_score.grade }}) 📊 ({{ productivity_score.label }})
15 **Overall Sentiment:** {{ sentiment.label | capitalize }}
16
17 ---
18
19 ## 📄 Executive Summary
20 {{ short_summary }}
21
22 ## 📄 Detailed Summary
23 {{ detailed_summary }}
24
25 ## 📄 Key Bullet Points
26 {% for point in bullet_points %}- {{ point }}
27 {% endfor %}
28
29 ---
30
31 ## 📄 Decisions Made
32 {% if decisions %}{% for d in decisions %} ✅ {{ d }}
33 {% endfor %}{% else %}_No formal decisions detected._
34 {% endif %}
35
36 ---
37
38 ## ✅ Action Items

```

Figure 4.8: Report Generation Code

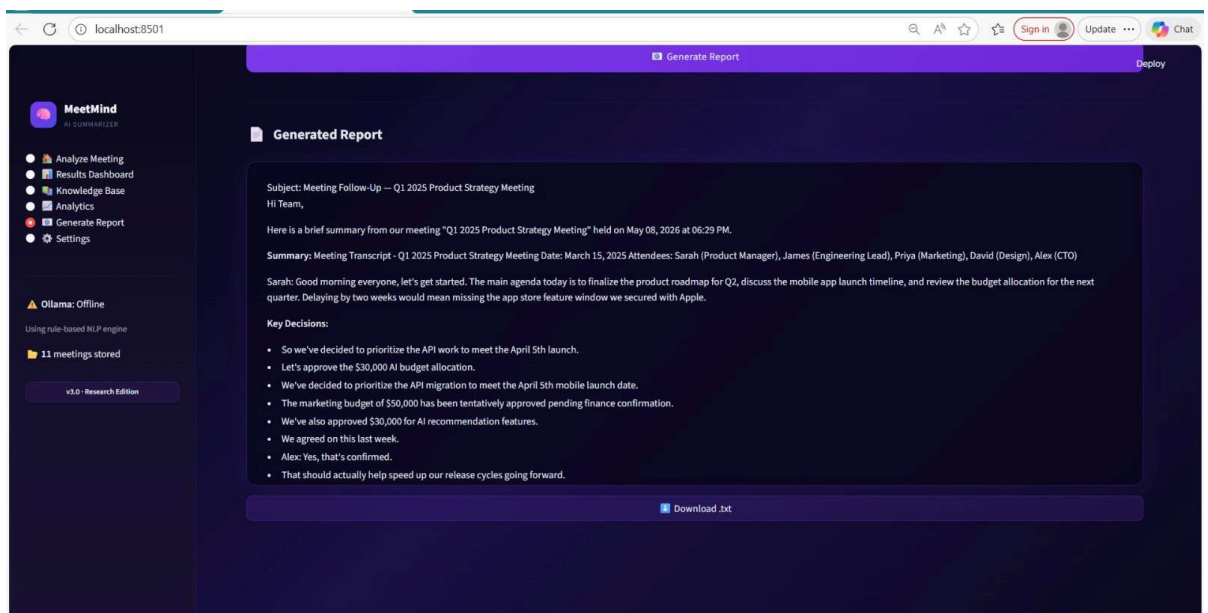


Figure 4.9: Generated Meeting Report

4.3. Overall Module Testing

After integrating all modules, complete system testing was performed.

Test Case	Expected Result	Status
Meeting Analysis	Summary Generation	Passed
Decision Detection	Correct Decision Extraction	Passed
Action Item Extraction	Task Identification	Passed
Risk Detection	Concern Identification	Passed
Knowledge Base Storage	Meeting Saved Successfully	Passed
Analytics Dashboard	Statistics Generated	Passed
Report Generation	Report Created Successfully	Passed
Report Download	Download Functionality Working	Passed

The testing results indicate that all modules functioned successfully and achieved the desired project objectives.

The implementation demonstrates the practical application of Artificial Intelligence, Natural Language Processing, Machine Learning, and Data Analytics techniques for intelligent meeting analysis, automated summarization, action item generation, and organizational knowledge management.

CHAPTER – 5

PROJECT STAGE-I OUTCOMES

5. PROJECT STAGE-I OUTCOMES

Project Stage-I focused on the planning, analysis, design, and initial implementation of the proposed system. During this phase, the requirements of the **AI Powered Meeting Summarizer and Action Item Generator** were identified and the core modules of the system were developed successfully.

The objective of Stage-I was to establish an intelligent meeting analysis platform capable of generating meeting summaries, extracting action items, identifying decisions, analyzing sentiments, and providing productivity insights. Various Artificial Intelligence, Natural Language Processing (NLP), and data analytics technologies were studied and implemented to achieve the desired functionality. The developed modules were tested individually to verify their performance and correctness.

5.1. Requirement Analysis Outcome

The first activity in Stage-I was the analysis of existing meeting management and documentation systems and identification of their limitations. Problems such as manual note-taking, missing action items, inefficient meeting tracking, and lack of knowledge management were studied.

Requirements Identified:

- Automatic meeting summarization
- Action item extraction

- Decision detection
- Risk identification
- Sentiment analysis
- Productivity scoring
- Knowledge base management
- Analytics dashboard
- Automated report generation

Outcome:

The requirement analysis confirmed that Artificial Intelligence and Natural Language Processing techniques can effectively automate meeting analysis and significantly reduce manual documentation efforts.

5.2. Literature Survey Outcome

A detailed literature survey was conducted to understand existing meeting summarization systems, Natural Language Processing techniques, and Large Language Models.

Outcome:

The survey identified TF-IDF, NLP-based text processing, sentiment analysis, and LLM-based summarization as suitable techniques for developing an intelligent meeting analysis system.

5.3. System Design Outcome

The overall architecture and workflow of the system were designed during Stage-I. The interaction between different modules such as meeting analysis,

knowledge base management, analytics dashboard, and report generation was defined.

Outcome:

A complete system architecture was successfully designed to support automated meeting analysis, information extraction, knowledge management, and report generation.

5.4. Dataset Preparation Outcome

A meeting transcript dataset was prepared for system testing and validation. Sample meeting conversations containing discussions, decisions, tasks, and participant information were collected and organized.

- Meeting transcript collection
- Dataset organization
- Meeting metadata preparation
- Participant information management
- Sample discussion preparation

Outcome:

A structured meeting transcript dataset was successfully prepared for testing summarization, action-item extraction, sentiment analysis, and report generation modules.

5.5. Module Development Outcome

The core modules of the project were implemented and tested successfully.

5.5.1. Meeting Analysis Module

The Meeting Analysis Module was developed using Python, spaCy, NLTK, and Scikit-Learn libraries.

Features Implemented:

- Transcript processing
- Automatic summarization
- Decision extraction
- Action item identification
- Risk detection
- Keyword extraction
- Sentiment analysis

Outcome:

The module successfully generated meeting summaries, extracted decisions, identified action items, and analyzed participant sentiments.

5.5.2. Knowledge Base and Analytics Module

The Knowledge Base and Analytics Module was developed using SQLite and Plotly visualization libraries.

Features Implemented:

- Meeting storage
- Search functionality
- Historical meeting retrieval
- Productivity analytics

- Trend analysis
- Dashboard generation

Outcome:

The module successfully stored meeting records and generated meaningful analytics and productivity insights.

5.5.3. Report Generation Module

The Report Generation Module was developed to create structured meeting reports automatically.

Features Implemented:

- Executive summary generation
- Detailed meeting reports
- Action item reports
- Decision reports
- Downloadable reports

Outcome:

The module successfully generated downloadable meeting reports containing summaries, decisions, tasks, and meeting insights.

5.6. Testing Outcome

Test Case	Expected Result	Status
Meeting Analysis	Summary Generation	Passed
Decision Detection	Correct Decision Extraction	Passed
Action Item Extraction	Task Identification	Passed
Risk Detection	Concern Identification	Passed
Knowledge Base Storage	Meeting Saved Successfully	Passed
Analytics Dashboard	Statistics Generated	Passed
Report Generation	Report Created Successfully	Passed
Report Download	Download Functionality Working	Passed

Outcome:

All modules performed according to the expected results and achieved the required functionality.

5.7. Achievements of Project Stage-I

- Problem analysis completed.
- Literature survey completed.
- System architecture designed.
- Meeting transcript dataset prepared.
- Meeting analysis module developed.
- Automatic summarization implemented.
- Decision extraction module implemented.
- Action item extraction module implemented.
- Sentiment analysis functionality implemented.
- Knowledge base module developed.
- Analytics dashboard created.
- Report generation module implemented.
- Module-level testing completed successfully.

CHAPTER – 6

CONCLUSION

6. CONCLUSION

The **AI Powered Meeting Summarizer and Action Item Generator** was successfully designed and developed to automate the process of meeting analysis and documentation. The system uses Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to analyze meeting transcripts and generate meaningful outputs such as summaries, action items, decisions, risks, and productivity insights.

The project was implemented using Python, Streamlit, spaCy, NLTK, SQLite, and other supporting technologies. The developed system can process meeting discussions efficiently and reduce the manual effort required to prepare meeting minutes. It also helps users track important decisions and assigned tasks, ensuring better accountability and follow-up.

The Knowledge Base Module allows storage and retrieval of meeting records, while the Analytics Dashboard provides useful insights regarding meeting productivity and participant contributions. The Report Generation Module further enhances usability by creating structured meeting reports automatically.

CHAPTER – 7

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7. REFERENCES

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